

1. Administrative & Study Identification

Full Study Title: A PHASE 3, RANDOMIZED, OPEN-LABEL TRIAL TO EVALUATE THE SAFETY, TOLERABILITY, AND IMMUNOGENICITY OF RESPIRATORY SYNCYTIAL VIRUS (RSV) PREFUSION F SUBUNIT VACCINE FORMULATED IN MULTIDOSE VIALS IN HEALTHY FEMALE ADULTS **Short Title/Acronym:** Amélie / RSVpreF MDV vs SDV Study **Protocol Number:** C4841001 **NCT Number:** NCT06473519 **Sponsor Name:** Pfizer **Investigational Product:** RSVpreF (Abrysvo) Bivalent RSV prefusion F protein subunit vaccine **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the immunogenicity, safety, and tolerability of the RSVpreF multidose vial (MDV) presentation compared with the single-dose vial (SDV) presentation at 1 month after vaccination. **Secondary Objectives:** To compare the safety profiles, frequency of local and systemic reactogenicity, and seroresponse rates between the MDV and SDV presentations.

2.2 Background & Rationale RSVpreF is currently available as a preservative-free, single-dose vial (SDV). RSVpreF offered in a multidose vial (MDV) with the preservative 2-phenoxyethanol (2-PE) may help reduce vaccination costs and decrease cold-chain storage capacity requirements, which is particularly beneficial in lower-middle-income countries (LMICs). This study aims to ensure that the MDV formulation provides noninferior immunogenicity and a comparable safety profile to the existing SDV formulation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (RSVpreF SDV) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 453 randomized participants (450 ultimately vaccinated) **Number of Sites:** 6-7 sites in the United States **Estimated Study Start:** 24-Jun-2024 **Estimated Primary Completion:** 20-Sep-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy, nonpregnant, nonbreastfeeding females 18 through 49 years of age at Visit 1 (Day 1). 2. Willing and able to comply with all scheduled visits, investigational plan, lifestyle considerations, and other study procedures. 3. Capable of giving signed informed consent.

4.2 Exclusion Criteria 1. Previous vaccination with any licensed or investigational RSV vaccine, or planned receipt of a nonstudy RSV vaccine throughout the study. 2. History of severe adverse reaction associated with a vaccine and/or severe allergic reaction (e.g., anaphylaxis) to any component of the vaccines being administered. 3. Immunocompromised participants with known or suspected immunodeficiency, or receipt of chronic systemic treatment with immunosuppressant medications within 60 days before enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single dose (approximately 0.5 mL) of RSVpreF administered either from a multidose vial (MDV with 2-PE preservative) or a single-dose vial (SDV without preservative). **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Single injection on Day 1; approximately 6 weeks of study participation follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications not affecting the immune response. **Prohibited:** Receipt of blood/plasma products or immunoglobulin within 60 days before study intervention or planned receipt throughout the study; immunosuppressants; other investigational or nonstudy RSV vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening/Baseline	Day 1 Informed Consent, Medical History, Blood Sample, Randomization, Single IM Vaccination Interim 1 Month post-dose Clinic Visit: Blood Sample for Immunogenicity Assessment End of Study Approx. Week 6 Final safety telephone call

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Noninferior immunogenicity of RSVpreF MDV compared with RSVpreF SDV at 1 month after vaccination, evaluated by the neutralizing geometric mean titres

(GMTs) for RSV A and RSV B. **Measurement Method:** Central Laboratory Analysis (Blood Samples)

7.2 Secondary & Exploratory Endpoints * Safety profile comparisons: Frequency of adverse events (AEs) and severe adverse events (SAEs). * Frequency and severity of local reactions (e.g., pain at injection site) and systemic events (e.g., fatigue). * Seroresponse rate differences for RSV A and RSV B.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient electronic diaries utilized to monitor solicited local/systemic reactions (e.g., pain, fatigue) post-vaccination. **Serious Adverse Events (SAEs):** Collected and monitored for the entire duration of the study (approx. 6 weeks). **Data Monitoring Committee (DMC):** Internal safety monitoring managed by Pfizer.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Evaluable immunogenicity population (\$n=434\$), Safety Set (\$n=450\$). **Sample Size Justification:** Powered to determine noninferiority where the lower 95% CI bound for the geometric mean ratio (GMR) of neutralizing GMTs (MDV to SDV) is \$> 0.67\$ (\$1.5\$-fold noninferiority margin) for both RSV A and B. **Handling of Missing Data:** Excluded from per-protocol immunogenicity analysis if significant blood sample protocol deviations occur.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite and remote monitoring of study data, interactive response technology system logs, and consent forms. **Audit Trail:** Details on eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: C4841001 **Registry Number:** NCT06473519 **Sponsor/Lead Organization:** Pfizer **Study Title:** Amélie **Official Regulatory Title:** A PHASE 3, RANDOMIZED, OPEN-LABEL TRIAL TO EVALUATE THE SAFETY, TOLERABILITY, AND IMMUNOGENICITY OF RESPIRATORY SYNCYTIAL VIRUS (RSV) PREFUSION F SUBUNIT VACCINE FORMULATED IN MULTIDOSE VIALS IN HEALTHY FEMALE ADULTS

12. Clinical Framework (RSV Vaccine Specific)

Primary Condition: Respiratory Syncytial Virus (RSV) Infection Prevention **Diagnosis Threshold:** N/A (Healthy Adults) **Medical Methodology:** Active Prophylactic Immunization (Vaccine) **Optimization Target:** Neutralizing antibody titer response for RSV A and RSV B

13. Study Procedures & Methodology

Management Pathway: Standard prophylactic vaccination administration pathway **Education Protocol (HCP):** Vaccine administration, multidose vial formulation handling, and cold-chain capacity management **Education Protocol (Patient):** Post-vaccination reactogenicity reporting and electronic diary training **Quality Audit Mechanism:** AE and SAE tracking, immunogenicity sample viability monitoring

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6 Follow-up weeks **Onsite Visit Cadence:** Day 1 and 1 Month Post-Vaccination **Remote Monitoring Frequency:** Final safety telephone call at end of study (~Week 6) **Checklist Review Interval:** Daily electronic diary review for 7 days post-vaccination

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				